

Ambient Systems

AI-Driven Compliance & Sustainability Solutions for the Built Environment

Company Overview

Ambient Systems is an AI-driven compliance and sustainability venture re-establishing its Singapore operations to pilot collaborations under BEAMP Cycle 6. Originally spun out of the National University of Singapore in 2021, we've evolved into a global innovation hub with strategic operations across three continents.

(HQ) – Core operations and system architecture

Hong Kong – Asia-Pacific market validation and R&D

Québec – North American research partnerships

Recognition & Patents

WIPO Patent

WO – AI Compliance System for construction verification

Panasonic Winner

Deep Tech Challenge 2023 – Innovation in built environment solutions

Google Cloud

-Up 2022 – AI deployment excellence

Business Activities

We develop AI systems for compliance automation, ESG intelligence, and performance verification across the construction and infrastructure sectors. Our solutions bridge the gap between regulatory requirements, operational excellence, and sustainability reporting.

Construction QA/QC

Automated quality assurance and control systems for building projects, ensuring compliance and performance standards are consistently met.

Infrastructure Digital Twins

Real-time virtual replicas of physical assets enabling predictive maintenance, optimization, and lifecycle management.

Sustainability Analytics

ESG reporting and green finance intelligence tools that track carbon metrics and support environmental compliance.

BEAMP Relevance & Singapore Focus

Built Environment Localization

Adapting our platform for Singapore's regulatory framework and construction standards.

Integrated Innovation

Combining FinTech, GreenTech, and ConTech modules for comprehensive solutions.

Automated Verification

AI-powered ESG reporting and compliance validation at scale.

Global Market Experience

1. **2021 – Singapore/ASEAN**
   Early pilots and academic collaborations through NUS
2. **2023 – Japan**
   Smart City Conference presentations at City Tech Tokyo
3. **2023 – Hong Kong/China**
   R&D expansion and market validation initiatives
4. **2024 – North America**
   Research partnerships and pilot deployments
5. **2024 – Europe**
   Market expansion and technology refinement

Products & Services

01

BEAMP Cycle 6 POC/Pilot Focus

Our proof-of-concept initiatives target three critical areas where AI can transform construction compliance and sustainability.

Concrete Quality – Automated verification of delivery documentation, slump tests, and cube-test data with real-time alerts

Construction Economics – QA performance metrics linked directly to financial milestones and project cost visibility

Sustainability Metrics – Embodied carbon tracking and materials documentation for ESG reporting requirements

02

Platform – ambientsystems.ai

A comprehensive digital ecosystem integrating compliance, analytics, and operational intelligence for the built environment.

Financial Metrics Integration – Connect quality assurance data with cost tracking and budget performance

Generative AI Planning – Scenario modeling for compliance workflows and logistics optimization

IoT Connectivity – Seamless sensor integration with automated data capture and validation

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Consulting – ivandamnjanovic.com

Specialized advisory services leveraging our patented technology and deep domain expertise.

- Civil Engineering performance audits and compliance reviews
- Energy systems optimization and carbon footprint analytics
- Controlled Environment Agriculture (CEA) digital twin development
- Custom IP and system design based on WO patent

Business Targets

BEAMP Cycle 6 represents a strategic inflection point for Ambient Systems—serving as both the anchor for our Singapore-based innovation hub and the launchpad for global deployment across construction, finance, and ESG markets.

Establish Singapore Entity

Launch new legal structure focused on AI compliance and ESG solutions tailored for the built environment sector.

Transfer IP Portfolio

Register complete intellectual property portfolio including WO patent under Singapore operations.

Complete

development of Ambient Verify and Ambient Twin platform components for commercial deployment.

Secure Pilot Partners

Execute LOIs and MOUs with pilot clients and ecosystem partners across construction and finance sectors.

Advance Technology Readiness

Progress from TRL 6 to TRL 9 through successful BEAMP pilot implementations and real-world validation.

: 2025–2028

2025 Q1-Q2

R&D Phase

Platform refinement, IP transfer, pilot recruitment

2025 Q3-Q4

Pilot Deployment

BEAMP implementations, data collection, validation

2026-2027

Commercial Scale

Market expansion, partnership development

2028

Global Deployment

Regional hubs, enterprise clients

Key Team Members

Ivan Damnjanovic

Founder & Systems Architect

Advertising & Communications